COST OF CAPITAL & FINANCIAL LEVERAGE

Quick Reference Formula Sheet

I. COST OF CAPITAL

1. Cost of Debt (Kd)

Irredeemable, before tax	**Kd = I / NP**
Irredeemable, after tax	**Kd = I(1 − t) / NP**
Redeemable, after tax	**Kd = [I(1−t) + (RV−NP)/n] ÷ [(RV+NP)/2]**

2. Cost of Preference Capital (Kp)

Irredeemable	**Kp = PD / NP**
Redeemable	**Kp = [PD + (RV−NP)/n] ÷ [(RV+NP)/2]**

3. Cost of Equity (Ke)

Dividend Price Method	**Ke = D1 / NP (or D / MP)**
Growth Model (Gordon)	**Ke = D1/NP + g ; g = b × r**
Earnings Price Method	**Ke = E / NP**
CAPM	**Ke = Rf + β (Rm − Rf)**
Realised Yield Method	**Based on actual dividends & growth received**

4. Cost of Retained Earnings (Kr)

Retained earnings	**Kr = Ke (1 − tp)(1 − b)**

5. Weighted Average Cost of Capital (WACC)

WACC	**WACC = Σ (Wi × Ki)**
Weights used	**Book value weights OR Market value weights**

6. Marginal Cost of Capital (MCC)

MCC	**Cost of the last / next additional rupee of capital raised**

II. FINANCIAL LEVERAGE

1. Operating Leverage (OL)

Operating Leverage	**OL = Contribution / EBIT**
Degree of Operating Lev.	**DOL = %Δ EBIT / %Δ Sales**

2. Financial Leverage (FL)

Financial Leverage	**FL = EBIT / EBT**
Degree of Financial Lev.	**DFL = %Δ EPS / %Δ EBIT**
EBT	**EBT = EBIT − Interest**

3. Combined Leverage (CL)

Combined Leverage	**CL = OL × FL = Contribution / EBT**
Degree of Combined Lev.	**DCL = %Δ EPS / %Δ Sales**

4. Earnings Per Share (EPS)

EPS	**EPS = [(EBIT − I)(1 − t) − PD] / N**

5. Indifference Point (EBIT–EPS Analysis)

Indifference Point	**[(EBIT−I1)(1−t)−PD1]/N1 = [(EBIT−I2)(1−t)−PD2]/N2**

6. Financial Break-Even Point

Financial BEP	**Financial BEP = I + PD / (1 − t)**

Exam Tips

- Use market value weights for WACC unless the question specifically states book value weights.
- DOL, DFL and DCL can each be computed either from the % change formula or the direct ratio formula — both must give the same answer; use the ratio formula to save time in numericals.
- At the indifference point, EPS is equal under both financing plans — beyond this EBIT level, the higher-leverage plan gives higher EPS.

Notation Used

I = Annual interest	**NP** = Net proceeds of security	**RV** = Redemption value	**n** = Years to maturity/redemption	**t** = Corporate tax rate
PD = Preference dividend	**D1** = Expected dividend per share	**MP** = Market price per share	**g** = Growth rate	**b** = Retention ratio

r = Return on retained earnings	**E =** Earnings per share	**Rf / Rm =** Risk-free / market return	**β =** Beta (systematic risk)	**tp =** Shareholder's personal tax rate
EBIT / EBT = Earnings before interest & tax / before tax	**N =** No. of equity shares			

Note: Notation/formula variants (e.g., MP vs NP, flotation cost treatment) can differ across syllabi — please verify against your institution's prescribed textbook, UGC curriculum, and board (ISC/CBSE) norms.